25900315

MAHATMA GANDHI UNIVERSITY KOTTAYAM
MGU-BCA (HONOURS) Regular EXAMINATION October 2025
Third SEMESTER
Core Course(CC)-MG3CCRBCA201-DATABASE MANAGEMENT SYSTEMS
(2024 ADMISSION ONWARDS)

Duration: 2 Hours Maximum Marks: 70

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Very Short Answer Questions
Answer all the questions
Each question carries 2 marks

1. Can a primary key be NULL or duplicate? Justify. [An]/[CO1]
   A Primary Key cannot be NULL because it must uniquely identify a tuple, and it cannot contain duplicates because that would prevent distinguishing between two different records.
2. Define TRC. [U]/[CO2]
   Tuple Relational Calculus (TRC) is a non-procedural query language that uses the formula { t | P(t) },t is the tuple variable and P(t) is a predicate.
3. State the reflexivity property of functional dependencies. [K]/[CO3]
   If $Y \subseteq X$, then $X \rightarrow Y$
4. Write the uses of Cassandra [A]/[CO4]
   Apache Cassandra is a highly scalable, distributed NoSQL database designed to handle massive volumes of data across many servers without any single point of failure.
5. What is the role of lock manager in concurrency control? [U] / [CO3]
   The Lock Manager is the DBMS component responsible for managing locks requested by concurrent transactions, maintaining a lock table to track granted locks, and forcing conflicting transactions to wait.

Part B

Short Answer Questions
Answer any 5 questions
Each question carries 6 marks

6. Analyze the purpose of DBMS in managing large amounts of data. Why is it preferred over file-based data storage? [An] / [CO1]

The purpose of a Database Management System (DBMS) is to provide an efficient and reliable way to **store, define, manipulate, and share** a large collection of related data for various applications, while ensuring the **safety and integrity** of the data.

Advantages of DBMS over File System

Redundancy Control, Data Consistency, Data Integrity,Data Security,Data Sharing, Concurrent Access Control,Efficient Data Access(Briefly explain any 4)

7. Summarize Serializability importance in transaction management. [U] / [CO3]

   In multi-user systems, many transactions run at the same time. Without careful control, interleaved operations could cause Lost Update, Temporary Update and Incorrect Summary.Serializability also ensuresACID Properties.(Explain)

8. Write SQL queries to demonstrate the use of logical operators (AND, OR, NOT) in filtering records. [A] / [CO2]

Operator	Purpose	Example Query
AND	Returns a row only if **both** connected conditions are TRUE.	SELECT Ename, Salary FROM Employee WHERE Dept_ID = 5 AND Salary > 50000;
OR	Returns a row if **at least one** of the connected conditions is TRUE.	SELECT CName, CLevel FROM Course WHERE CLevel = 'UG' OR CLevel = 'PG';
NOT	Negates the condition; returns a row if the condition is **FALSE**.	SELECT * FROM Project WHERE NOT PLocation = 'New Delhi';

9. Compare column family and graph data models of NoSQL databases.[An] / [CO4]

   <u>Column family</u>

   Data is organized into rows, where each row contains a key and a set of column families. Columns are flexible and can be sparse (a row may not have all columns).

   Use Cases-Time-series data, high-volume logging, real-time analytics, and applications

requiring extreme horizontal scaling and write-heavy operations.
Excellent horizontal scalability (sharding/distribution is by row key).

Graph Data Model

Data is structured as a network of **Nodes** (entities) and **Edges** (relationships) with associated properties.
Use cases-Social networks, recommendation engines, fraud detection, and any system where relationships are as important as the data itself.
Scales well for large numbers of nodes and edges; performance depends heavily on the complexity of the traversal.

10. Discuss about dirty read, nonrepeatable read and phantom problem in a transaction.
[U] / [CO3]

Dirty Read (Temporary Update Problem):

a. Occurs when: Transaction T1 reads a data item that has been modified by a concurrent transaction T2, but T2 has not yet committed.
b. Problem: If T2 later rolls back its changes, T1 has read a value that never officially existed in the database, leading to inconsistent results.

Nonrepeatable Read:

c. Occurs when: Transaction T1 reads the same data item twice. Between the two reads, a different transaction T2 updates or deletes that specific item and commits.
d. Problem: T1 gets different values for the same item upon repeating its read, violating the transaction's isolation.

Phantom Problem (Phantom Read):

e. Occurs when: Transaction T1 executes a query (e.g., a SELECT with a WHERE clause) that retrieves a set of rows. A different transaction T2 then inserts a new row that satisfies T1's WHERE clause and commits.
f. Problem: When T1 repeats the query, it sees a new, unexpected "phantom" row, changing the *set* of rows retrieved

11. Explain the concept of FULL OUTER JOIN in SQL with proper syntax, example tables, and expected output. Also, discuss how it differs from LEFT and RIGHT OUTER JOINS. [U] / [CO2]

FULL OUTER JOIN in SQL

- **Concept:** The FULL OUTER JOIN operation returns the **union of all rows** from both the Left table and the Right table. It includes:
    1. All matching rows from both tables.
    2. All un-matched rows from the Left table, padded with **NULL** values for the Right table's columns.
    3. All un-matched rows from the Right table, padded with **NULL** values for the Left table's columns.

Syntax:

```
SELECT A.*, B.*
FROM TableA AS A
FULL OUTER JOIN TableB AS B ON A.Key = B.Key;
```

(Explanation 3marks, Example-3 marks)

Difference from LEFT and RIGHT OUTER JOINs:

- FULL OUTER JOIN preserves *all* data from *both* tables
- LEFT OUTER JOIN prioritizes the Left table, returning all its rows and only the matched rows from the Right table.
- RIGHT OUTER JOIN prioritizes the Right table, returning all its rows and only the matched rows from the Left table.

12. Analyze the importance of primary keys in the relational model with relevant examples. [An] / [CO2]

    A PK is a minimal set of attributes that **uniquely identifies** every single tuple (row) in the table. This guarantees that each record is distinct and can be referenced without ambiguity, which is a core tenet of the relational model.

    (Explain with example)

Part C

Essay Questions

Answer any 2 questions

Each question carries 15 marks

13. Explain the notations of ER diagram.Create an Entity-Relationship (ER) diagram for a university database that includes entities like 'student','course', and 'instructor'. Define suitable relationships and attributes for each entity. [A] / [CO2]

Concept	Notation	Description
Entity Type	Rectangle	Represents a collection of real-world objects (e.g., Student, Course).
Attribute	Ellipse	Represents a property of an entity (e.g., Name, ID).
Key Attribute	Underlined Ellipse	An attribute that is the Primary Key for the entity (e.g., SId).

Relationship Type	Diamond	Represents an association between two or more entities (e.g., TEACHES).
Total Participation	Double Line	Indicates that every entity instance must participate in the relationship.

(Notations-5marks, Diagram-5 marks,Relationships and entities-5 marks

14. Define a data model. Explain the different types of data models in DBMS with diagrams and examples. [K] / [CO1]

A **data model** is a collection of concepts and rules used to describe the **structure of a database**.
Different Data Models are
Relational Model
E-R model
Hierarchical Model
Network model
Object oriented model.

15. With suitable steps and examples, explain how to compute the closure of a set of functional dependencies. [A] / [CO3]

Given a relation schema R, a set of functional dependencies F, and a set of attributes $X \subseteq R$, the closure $X^+$is defined as: The set of all attributes that are functionally determined by X using the dependencies in F.

compute X^+ by iteratively applying Armstrong's Axioms (reflexivity, augmentation, transitivity) until no more attributes can be added (Value liberally)

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